Graph-Based Molecular Generation via Adversarial Learning with Hierarchical Discriminators and Reward-Driven Objectives

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***Abstract*—Recent advances in deep-generative models have allowed the generation of chemically valid and novel molecules with desirable properties for further investigation. In this work, we provide such a model, integrating generative adversarial networks (GANs), hierarchical graph discrimination, and reinforcement learning. Our graph-based generator generates molecular graphs in the form of adjacency and node feature matrices. To improve robustness and stability during training, we implement a hierarchical differentiable pooling (DIFFPOOL)-based discriminator. This type of discriminator allows our model to learn coarse-grained graph representations during training. Additionally, we introduce a reinforcement learning component in which a policy network receives reward signals based on predicted chemical properties, guiding the generator toward structures with high drug-likeness and other target metrics. In evaluating our approach, we look for validity, uniqueness, novelty, and alignment with drug-likeness metric scores. The results demonstrate improved generation quality and training stability compared to baseline graph GANs, particularly when incorporating DIFFPOOL and reinforcement-guided reward shaping. Our findings suggest that the combination of hierarchical discrimination with reinforcement learning shows a positive trajectory for property-driven molecular design in drug discovery and materials science.**

***Index Terms*—GAN, Hierarchical discriminator, DIFFPOOL, Reinforcement learning, Validity, Uniqueness, Novelty, Drug-likeness**

Nomenclature

$A \in \mathbb{R}^{nxn}$	adjacency matrix
$T = [t_1, t_2, ..., t_n] \in \mathbb{Z}^n$	atomic numbers (node features)
$B \in \mathbb{Z}^{nxn}$	bond order matrix (optional)
$G = (V, E)$	molecular graph with vertices V and edges E

I. Introduction

In our research, we used the QM9 dataset to train a GAN generator working alongside a hierarchical discriminator to generate chemically valid and novel molecules for scientific study. We used two different versions of the hierarchical discriminator, one with DIFFPOOL, and one without. In association with the generator and discriminator, we also implement a reward policy network to encourage the model's generation. The dataset we used, QM9, contained the following variables: mol_id, smiles, A, B, C, mu, alpha, homo, lumo, gap, r2, zpve, u0, u298, h298, g298, cv, u0_atom, u298_atom, h298_atom, g298_atom, Morgan_fingerprint, adj_matrix, node_features, and node_count. The variables that are significant for this work are adj_matrix, node_features, and node_count. These variables define the molecules that are fed into the generator during training. The adjacency matrix, adj_matrix, contains information on which bonds are present inside the molecule. The node features, node_features, contain information on which atoms are present within the molecule. In this study, we analyze the validity, novelty, uniqueness, and drug-likeness of molecules produced by the generator. The validity is based on whether or not the generated molecule is chemically valid, uniqueness is based on similarity of the generated molecule to previous molecules, novelty is determined by whether or not a similar molecule is already present within the training data, and drug-likeness is concerned with the similarity of the molecule generated to common drug-like molecules.

II. Literary Review

III. Methods

A. Adjacency Matrix to Molecule

1) Vertex Set (Atoms): Each atom is represented by a vertex:

$$V = \{v_i = Atom(t_i) | i = 1, 2, ..., n\} \quad (1)$$

Optionally, if chirality tags are given:

$$Chiral(v_i) = c_i \; for \; i \in C \subseteq \{1, ..., n\} \tag{2}$$

2) Edge Set (Bonds): For all i < j:

$$\text{If } A_{ij} > \tau : \quad o_{ij} = \begin{cases} B_{ij}, & \text{if } B_{ij} > 0 \\ 1, & \text{otherwise} \end{cases}$$
$$E \ni e_{ij} = (v_i, v_j, \beta(o_{ij})) \tag{3}$$

Where $\beta(o)$ maps bond order to bond type:

$$\beta(o) = \begin{cases} \text{SINGLE}, & o = 1 \\ \text{DOUBLE}, & o = 2 \\ \text{TRIPLE}, & o = 3 \\ \text{AROMATIC}, & o = 4 \\ \text{undefined}, & \text{otherwise} \end{cases} \tag{4}$$

3) Isolated Hydrogen Handling: Let:

$$H = i | t_i = 1 \wedge deg(v_i) = 0 \tag{5}$$

For each h $\in$ H:

$$j^* = \arg\max_{\substack{j \notin H \\ j \neq h}} A_{hj} \tag{6}$$

If $deg(v_{j^*}) <$ Valence$_{max} t_{j^*}$, then:

$$E \ni e_{hj^*} = (v_h, v_{j^*}, \text{SINGLE}) \tag{7}$$

Else:

$$V \leftarrow V \setminus \{v_h\} \tag{8}$$

4) Molecule: The final molecule is:

$$M = Sanitize(G = (V,\ E)) \tag{10}$$

B. Reward Policy Network

1) Validity Reward: Let M be a candidate molecule derived from (A, T). Let $\text{Valence}_{\max}(t_i)$ be the maximum valence for atom type t_i, and let $\deg(v_i)$ be the current explicit valence (degree) of atom $v_i \in V$.

IV. Ease of Use

A. Maintaining the Integrity of the Specifications

The IEEEtran class file is used to format your paper and style the text. All margins, column widths, line spaces, and text fonts are prescribed; please do not alter them. You may note peculiarities. For example, the head margin measures proportionately more than is customary. This measurement and others are deliberate, using specifications that anticipate your paper as one part of the entire proceedings, and not as an independent document. Please do not revise any of the current designations.

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Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

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C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{12}$$

$$R_{\text{validity}}(M) = \begin{cases} -0.5, & \text{if } M \text{ is None or fails sanitization} \\ \max\left(0,\ 5 - \left|\sum_{v_i \in V} (\text{Valence}_{\max}(t_i) - \deg(v_i))\right|\right), & \text{otherwise} \end{cases} \quad (11)$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use "(**??**)", not "Eq. (**??**)" or "equation (**??**)", except at the beginning of a sentence: "Equation (**??**) is . . ."

D. LaTeX-Specific Advice

Please use "soft" (e.g., `\eqref{Eq}`) cross references instead of "hard" references (e.g., `(1)`). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

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E. Some Common Mistakes

- The word "data" is plural, not singular.
- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited, such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)
- A graph within a graph is an "inset", not an "insert". The word alternatively is preferred to the word "alternately" (unless you really mean something that alternates).
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- In your paper title, if the words "that uses" can accurately replace the word "using", capitalize the "u"; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones "affect" and "effect", "complement" and "compliment", "discreet" and "discrete", "principal" and "principle".
- Do not confuse "imply" and "infer".
- The prefix "non" is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the "et" in the Latin abbreviation "et al.".
- The abbreviation "i.e." means "that is", and the abbreviation "e.g." means "for example".

An excellent style manual for science writers is [**?**].

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The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

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Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

Component heads identify the different components of your paper and are not topically subordinate to each other. Examples include Acknowledgments and References and, for these, the correct style to use is "Heading 5". Use "figure caption" for your Figure captions, and "table head" for your table title. Run-in heads, such as "Abstract", will require you to apply a style (in this case, italic) in addition to the style provided by the drop down menu to differentiate the head from the text.

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a) Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation "Fig. **??**", even at the beginning of a sentence.

TABLE I: Table Type Styles

Table Head	**Table Column Head**		
	Table column subhead	***Subhead***	***Subhead***
copy	More table copy[a]		

[a]Sample of a Table footnote.

Fig. 1: Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity "Magnetization", or "Magnetization, M", not just "M". If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization $\{A[m(1)]\}$", not just "A/m". Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)", not "Temperature/K".

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The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

References

Please number citations consecutively within brackets [**?**]. The sentence punctuation follows the bracket [**?**]. Refer simply to the reference number, as in [**?**]—do not use "Ref. [**?**]" or "reference [**?**]" except at the beginning of a sentence: "Reference [**?**] was the first ..."

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

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For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [**?**].

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